A Data Analysis Portfolio Project Using SQL

SQL PROJECT ON PIZZA SALES

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PROJECT OVERVIEW

Analyze pizza sales data to generate business insights

Tools Used: SQL (Pg Admin)

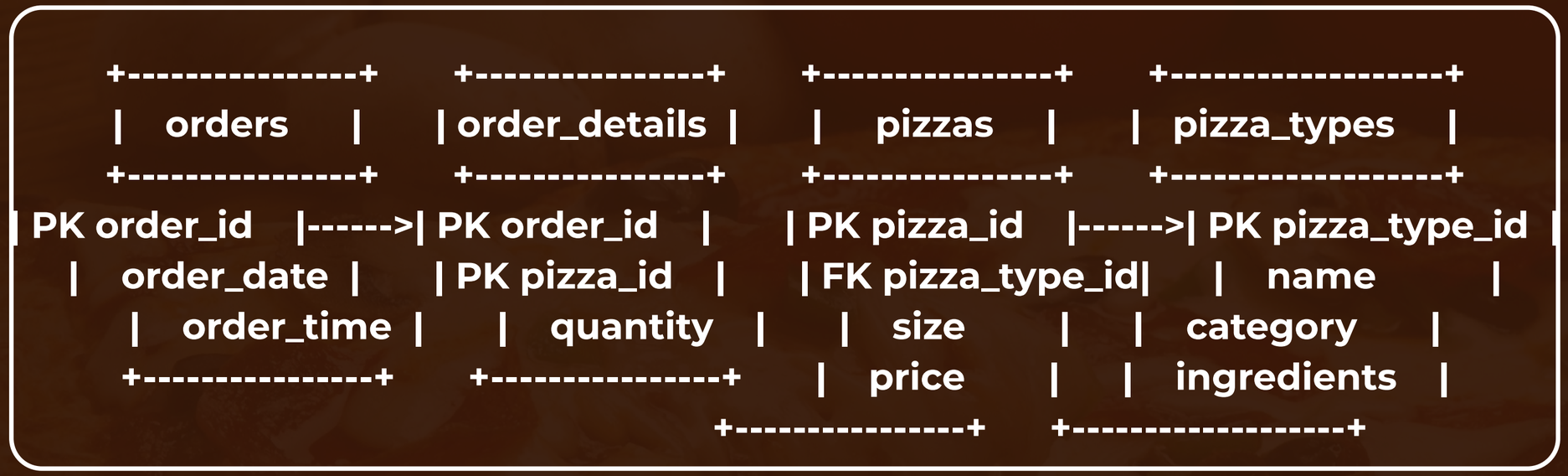
Database Tables:

- orders
- order_details
- pizzas
- pizza_types

Key Insights:

- Sales metrics
- Product trends
- Time-based patterns
- Revenue analysis

DATABASE SCHEMA DIAGRAM



KEY METRICS

Total Orders:

```
SELECT COUNT(order_id) AS total_order FROM orders;
```

Total Revenue:

```
SELECT SUM(od.quantity * p.price) AS total_revenue
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id;
```

Average Pizzas Ordered Per Day:

```
SELECT ROUND(AVG(quantity),0)
FROM (SELECT orders.order_date, SUM(order_details.quantity) AS quantity
    FROM orders JOIN order_details
    ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

PRODUCT ANALYSIS

Highest-Priced Pizza:

```
SELECT * FROM pizzas ORDER BY price DESC LIMIT 1;
```

Most Common Pizza Size Ordered:

```
SELECT p.size, SUM(od.quantity) AS total_quantity
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
GROUP BY p.size
ORDER BY total_quantity DESC LIMIT 1;
```

	size text	total_quantity bigint
1	L	18956

Top 5 Most Ordered Pizza Types:

```
SELECT pt.name, SUM(od.quantity) AS total_quantity
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY total_quantity DESC LIMIT 5;
```

	name text	total_quantity bigint
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

REVENUE ANALYSIS

```
WITH revenue_per_type AS (
    SELECT pt.name, SUM(od.quantity*p.price) AS revenue
    FROM order_details od
    JOIN pizzas p ON od.pizza_id = p.pizza_id
    JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
    GROUP BY pt.name
),
total AS (SELECT SUM(revenue) AS total_revenue FROM revenue_per_type)
SELECT rpt.name, rpt.revenue,
       ROUND((rpt.revenue/t.total_revenue*100),2) AS percentage
FROM revenue_per_type rpt, total t
ORDER BY rpt.revenue DESC;
```

Cumulative Revenue Over Time:

```
WITH daily_revenue AS (
    SELECT o.order_date, SUM(od.quantity * p.price) AS revenue
    FROM orders o
    JOIN order_details od ON o.order_id = od.order_id
    JOIN pizzas p ON od.pizza_id = p.pizza_id
    GROUP BY o.order_date
)
SELECT order_date, revenue,
       SUM(revenue) OVER (ORDER BY order_date) AS cumulative_revenue
FROM daily_revenue
ORDER BY order_date;
```

CATEGORY ANALYSIS

Category-wise Distribution:

```
SELECT pt.category, SUM(od.quantity) AS total_quantity
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY total_quantity DESC;
```

Top 3 Pizza Types by Revenue per Category:

```
WITH revenue_per_type AS (
    SELECT pt.category, pt.name, SUM(od.quantity * p.price) AS revenue
    FROM order_details od
    JOIN pizzas p ON od.pizza_id = p.pizza_id
    JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
    GROUP BY pt.category, pt.name
),
ranked AS (
    SELECT *, RANK() OVER (PARTITION BY category ORDER BY revenue DESC) AS rank
    FROM revenue_per_type
)
SELECT category, name, revenue
FROM ranked
WHERE rank <= 3
ORDER BY category, revenue DESC;
```

	category text	name text	revenue numeric
1	Chicken	The Thai Chicken Pizza	43434.25
2	Chicken	The Barbecue Chicken Pizza	42768.00
3	Chicken	The California Chicken Pizza	41409.50
4	Classic	The Classic Deluxe Pizza	38180.50
5	Classic	The Hawaiian Pizza	32273.25
6	Classic	The Pepperoni Pizza	30161.75
7	Supreme	The Spicy Italian Pizza	34831.25
8	Supreme	The Italian Supreme Pizza	33476.75
9	Supreme	The Sicilian Pizza	30940.50
10	Veggie	The Four Cheese Pizza	32265.70
11	Veggie	The Mexicana Pizza	26780.75
12	Veggie	The Five Cheese Pizza	26066.50

TIME-BASED ANALYSIS

Order Distribution by Hour:

```
SELECT EXTRACT(HOUR FROM order_time)
AS hour, COUNT(*) AS total_orders
FROM orders
GROUP BY hour
ORDER BY hour;
```

	hour numeric	total_orders bigint
1	9	1
2	10	8
3	11	1231
4	12	2520
5	13	2455
6	14	1472
7	15	1468
8	16	1920
9	17	2336
10	18	2399
11	19	2009
12	20	1642

KEY FINDINGS

- Revenue Drivers: Top performing pizza types and categories
- Peak Hours: When most orders occur
- Size Preference: Most popular pizza size
- Trends: Revenue growth over time
- Category Performance: Comparison between pizza categories

BUSINESS RECOMMENDATIONS

- Focus marketing on top-performing pizza types
- Optimize staffing during peak order hours
- Consider promotions for underperforming categories
- Monitor size preferences for inventory planning
- Use revenue trends for sales forecasting

Conclusion & Next Steps

Summary:

Comprehensive analysis of pizza sales data

Next Steps:

- Expand analysis with customer segmentation
- Incorporate cost data for profitability analysis
- Develop predictive models for demand forecasting

Thank You!

Questions?

Pizza Resto Presentation

THANK YOU FOR ATTENTION